Unconditional Proofs of Two Conjectures in OEIS A217564

With a Generalization to Prime-Counting Intervals

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Abstract

Let p_n be the nth prime, let $\pi(x)$ denote the number of primes not exceeding x, and put $a_d(n) = \pi(p_{n+1}/d) - \pi(p_n/d)$ for a fixed integer $d \geq 2$. We prove unconditionally that $\limsup_{n\to\infty} a_d(n) = \infty$ and that

$$\liminf_{N\to\infty} \frac{1}{N} \#\{1 \leq n \leq N : a_d(n) = 0\} \geq 1 - \frac{1}{d}.$$

For $d = 2$ these establish the two assertions recorded as conjectures in OEIS A217564. The zero-density bound follows from telescoping and the prime number theorem. For unboundedness, we combine the Maynard–Tao theorem for admissible linear forms with a Chinese-remainder construction of prime-free intervals surrounding multiples of a cluster of primes. The arguments are deductions from established results; no claim of priority or novelty of these deductions is made.

Keywords: prime gaps; Maynard–Tao theorem; Chinese remainder theorem; A217564; prime-counting function.
OEIS entry: A217564.
MSC 2020: 11N05, 11N36, 11A07.

1 Statement of the results

The Online Encyclopedia of Integer Sequences (OEIS) defines A217564 by

$$a(n) = \pi\left(\frac{p_{n+1}}{2}\right) - \pi\left(\frac{p_n}{2}\right). \tag{1}$$

The entry records the conjectures that $a(n)$ is unbounded and that 0 appears infinitely often [1]. We prove stronger statements simultaneously for all fixed integer scaling factors $d \geq 2$.

Theorem 1. *For every fixed integer $d \geq 2$, define*

$$a_d(n) = \pi\left(\frac{p_{n+1}}{d}\right) - \pi\left(\frac{p_n}{d}\right) \qquad (n \geq 1).$$

Then

$$\limsup_{n\to\infty} a_d(n) = \infty, \tag{2}$$

$$\liminf_{N\to\infty} \frac{\#\{1 \leq n \leq N : a_d(n) = 0\}}{N} \geq 1 - \frac{1}{d}. \tag{3}$$

In particular, both conjectures recorded in OEIS A217564 hold unconditionally on setting $d = 2$.

2 A positive lower density of zeros

Proof of (3). As the values $a_d(n)$ are nonnegative integers, the number of indices with $a_d(n) > 0$ is at most their sum. By telescoping and $\pi(2/d) = 0$,

$$\begin{aligned}\#\{1 \leq n \leq N : a_d(n) > 0\} &\leq \sum_{n=1}^{N} a_d(n) \\ &= \pi\left(\frac{p_{N+1}}{d}\right) - \pi\left(\frac{2}{d}\right) = \pi\left(\frac{p_{N+1}}{d}\right).\end{aligned}$$

The prime number theorem, together with $\pi(p_{N+1}) = N + 1$, yields

$$\pi\left(\frac{p_{N+1}}{d}\right) \sim \frac{p_{N+1}}{d \log(p_{N+1}/d)} \sim \frac{N}{d} \qquad (N \to \infty),$$

where d is fixed. Consequently,

$$\frac{\#\{1 \le n \le N : a_d(n) = 0\}}{N} \ge 1 - \frac{\pi(p_{N+1}/d)}{N} = 1 - \frac{1}{d} + o(1),$$

which proves (3). □

3 Prime clusters inside dilated prime gaps

We use the following established formulation of the Maynard–Tao theorem for linear forms [3, 2].

Theorem 2 (Maynard–Tao, quoted)**.** *For each integer $m \ge 2$ there exists k_m such that, whenever $k \ge k_m$ and $L_1(u), \dots, L_k(u)$ are pairwise nonproportional, admissible integer linear forms with positive leading coefficients, at least m of $L_1(u), \dots, L_k(u)$ are prime for infinitely many positive integers u. Here admissibility means that for every prime q, some residue class $u \pmod q$ makes every $L_j(u)$ nonzero modulo q.*

Lemma 3. *Fix integers $d, m \ge 2$. There exist integers $W > 0$, b and distinct nonnegative integers $h_1, \dots, h_k$, with $H = \max_j h_j$, such that:*

(i) the linear forms $L_j(u) = Wu + b + h_j$ are admissible;

(ii) for every sufficiently large $u \in \mathbb{N}$, every integer in $[d(Wu + b), d(Wu + b + H)]$ is composite;

(iii) for infinitely many u, at least m values among $L_j(u)$ are prime.

Proof. Choose $k \ge k_m$ as in Theorem 2 and put

$$D = d\,k!, \qquad h_j = (j-1)D \ (1 \le j \le k), \qquad H = (k-1)D.$$

Consider the finite set

$$T = \{t \in \mathbb{Z} : 1 \le t < dH,\ d \nmid t\}.$$

For each $t \in T$, choose a different prime $\ell_t > dH$. Let

$$W = d \prod_{t \in T} \ell_t.$$

By the Chinese remainder theorem, choose an integer b satisfying

$$b \equiv 1 \pmod d, \qquad db + t \equiv 0 \pmod{\ell_t} \quad (t \in T). \tag{4}$$

The moduli d and ℓ_t are pairwise coprime. Fix such a b and define $L_j(u) = Wu + b + h_j$.

Admissibility. Let q be any prime. If $q \mid d$, then $L_j(u) \equiv 1 \pmod q$ for every u, j. If $q = \ell_t$ for some $t \in T$, multiply by d and use (4):

$$dL_j(u) \equiv dh_j - t \pmod{\ell_t}.$$

The integer $dh_j - t$ is nonzero, because $d \mid dh_j$ but $d \nmid t$, and its absolute value is strictly less than $dH < \ell_t$. Hence $L_j(u) \not\equiv 0 \pmod{\ell_t}$ for all u, j. Finally, suppose $q \nmid W$. If $q \le k$, then $q \mid k!$ and all $h_j \equiv 0 \pmod q$; thus the k forms have the same single root modulo q. If $q > k$, they have at most $k < q$ roots altogether. In either case some residue class avoids all roots, proving (i).

A prime-free interval. Fix u large enough that the integers in the interval in (ii) exceed W, d and all the ℓ_t. Every integer in this interval has the form

$$d(Wu + b) + t, \qquad 0 \le t \le dH.$$

If $d \mid t$ it is divisible by d and is composite. Otherwise $t \in T$ and it is divisible by ℓ_t, by (4) and $\ell_t \mid W$; it is again composite. This proves (ii).

The forms L_j are pairwise nonproportional (they have equal leading coefficient W and distinct constants) and have positive leading coefficient, so Theorem 2 applies and proves (iii). □

Proof of (2). Fix $d \geq 2$ and an arbitrary $m \geq 2$. Choose the forms and interval from Lemma 3. For infinitely many arbitrarily large u, at least m distinct numbers $L_j(u)$ are prime, and the interval

$$I_u = [d(Wu + b), d(Wu + b + H)]$$

consists entirely of composite integers. Let $P < Q$ be the consecutive primes immediately surrounding I_u; thus $P < \min I_u \leq \max I_u < Q$. All m prime values $L_j(u)$ lie between $Wu + b$ and $Wu + b + H$, so they lie strictly between P/d and Q/d. Writing $P = p_n$, $Q = p_{n+1}$ gives

$$a_d(n) = \pi(Q/d) - \pi(P/d) \geq m.$$

As the selected u increase without bound, so do the enclosing prime gaps and their indices n. Therefore $a_d(n) \geq m$ holds for infinitely many n. Since $m \geq 2$ was arbitrary, (2) follows. □

4 Consequences and scope

Corollary 4 (OEIS A217564)**.** *For the sequence (1),*

$$\limsup_{n\to\infty} a(n) = \infty \qquad \textit{and} \qquad \liminf_{N\to\infty} \frac{\#\{n \leq N : a(n) = 0\}}{N} \geq \frac{1}{2}.$$

Thus $a(n) = 0$ infinitely often and $a(n)$ is unbounded.

Scope and attribution. The OEIS entry records the two conclusions as conjectures [1]. The analytic inputs in this note are the established prime number theorem and Maynard–Tao theorem. The CRT argument is supplied in detail to make the deduction checkable. These statements are unconditional, but this note does not assert that the application or generalization is previously unpublished; priority requires an independent literature assessment.

AI authorship and verification disclosure

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